

NEURAL SUBSYSTEM (Generative & Semantic Reasoning)

Probabilistic Natural Language Processing & Linguistic Formalization (Zero Physics Calculations)

Natural Language Interpretation

Ingests unstructured operator requests with variable phrasing, arbitrary syntax, and high-level routing intents.

Context Disambiguation via Optical RAG

Extracts localized k -hop subtopology G_{sub} and binds active ITU-T grid boundaries to scope LLM context.

Formal PDDL Translation

Acts strictly as a semantic compiler translating linguistic intent into typed PDDL goal predicates S_{PDDL} .

Semantic Agreement Judge

Evaluates reconstructed intent I_{recon} against original I_{NL} to compute semantic divergence d_{sem} .

FORMAL NEUROSymbolic BOUNDARY CONTRACT

Strict Type Hierarchy & Predicates (avoid-node, avoid-link, min-gsnr, max-hops)

Context-Free Grammar (CFG) Structural AST Audit: $v_{\text{struct}} \in \{0, 1\}$

SYMBOLIC SUBSYSTEM (Deterministic Solvers & Physical Simulation)

Topological Pruning, Combinatorial Path Search & Analytical Coherent GN-Model Physics (Zero Hallucination)

Topological Graph Pruning

Filters G_{sub} by eliminating vertices and links designated by avoid-predicates: $\hat{G}_{\text{sub}} = (\hat{V}_{\text{sub}}, \hat{E}_{\text{sub}})$.

Yen's K-Shortest Paths

Computes loopless candidate lightpaths over $\hat{G}_{\text{sub}}$, enforcing max-hop cardinality bounds $|\pi| \leq h_{\text{max}}$.

Coherent GN-Model Engine

Calculates span ASE noise accumulation, Kerr Non-Linear Interference (NLI), GSNR, and received power P_{rx} .

Risk-Adaptive Decision Gate

Evaluates $D(U_{\text{sem}}, \text{QoT}_{\text{valid}})$ to trigger autonomous auto-approval, physical replan, or early clarify.